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NPS/001/008 – Technical Specification for Gapless Metal Oxide Surge Arrestors

1. Purpose

The purpose of this document is to detail the technical requirements for gapless metal oxide surge arrestors used on the 11kV, 20kV, 33kV, 66kV and 132kV overhead line distribution networks of CE Electric UK

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
NPS/001/008	1.0	Technical Specification for Gapless Metal Oxide Surge Arrestors
DSS/007/018	1.0	Specification for Surge Arrestors
MS/O-37b	2003	Technical Specification for Gapless Metal Oxide Surge Arrestors

2. Scope

This document applies to non-linear gapless metal oxide surge arrestors without spark gaps designed to limit voltage surges on the 11kV, 20kV, 33kV, 66kV and 132kV overhead distribution network of CE Electric UK.

Appendix 1 of this specification contains self certification conformance declarations that shall be completed by the supplier for each product offered. Appendix 2 shall also be completed with the relevant technical information.

3. Technical Requirements

3.1 General

Gapless metal oxide surge arrestors shall be designed and tested in accordance with BS EN 60099 Part 4; 2006. The ranges of products include 11kV, 20kV and 33kV pole mounted stand-off arrestors and pedestal type arrestors for 66kV and 132kV arrangements. In all instances surge arresters will be used as a means of mechanical support for a cable termination, overhead line jumper or bus bar arrangement.

3.2 Service Conditions

The arrestor shall be suitable for operation under normal service conditions as detailed in BS EN 60099 – 4 Clause 5.4.1 with the addition of the following: -

Condition	Value	Information from -
Wind Speed	34 m/s	BS EN 62271-1:2008
Ice Loading	10mm radial	ENA TS 41-36 Issue 2
Rainfall	3mm/minute	IEC 60060 – 1 1998 Section 9.2 Table 2
Orientation	Vertical or horizontal	Company requirement

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3.3 System Parameters

Surge arrestors shall compatible with the system parameters detailed in the table below.

Rated Voltage	11kV	20kV	33kV	66kV	132kV
Maximum System Voltage (kV rms)	12	24	36	72.5	145
Rated Frequency (Hz)	50	50	50	50	50
Rated Current (A)	630	630	800	800	2000
Maximum Fault Currents					
phase-phase (kA rms)	13.1	10.1	20	20	25
phase-earth (kA rms)	4	4.5	4.5	3.5	20
Fault Duration					
phase-phase (seconds)	3.0	3.0	3.0	3.0	3.0
phase-earth (seconds)	10	10	10	10	3
Maximum Earth Fault Factor	1.73	1.73	1.73	1.73	1.4
Lightning Impulse Withstand Level – (1,25/50 μ s) kV peak	95	125	170	325	650
Power Frequency Voltage – Wet Withstand (kV)	45	70	90	140	275
Creepage Distance (mm) (25kV/mm)	300	600	900	1815	3625
Flashover Distance (mm) Basic electrical Clearance Phase – Earth (BS 7354 table 4)	160	220	320	630	1500
Short Circuit Current Rating (MVA)	250	350	1145	2286	5700

3.4 Arrestor Performance

The arrestors shall meet the performance requirements in accordance with BS EN 60099 Part 4 as summarised in the table below.

Surge arrestor Performance	11kV	20kV	33kV	66kV	132kV
Line Discharge Class (BS EN 60099 – 4)	1	1	2	2	2
Nominal Discharge Current (kA)	10	10	10	10	10
High Current Impulse (kA)	65	65	100	100	100
Highest System Voltage Rating (kV)	15.0	24.0	36.0	72.0	145.0
Continuous Voltage (kV)	12.0	19.2	28.8	57.6	116
Over voltage Capability (kV for 1 second)	17.1	27.4	41.4	82	140
Rated Short Circuit Current – Outdoor Application (kA)	16	16	25	25	25

3.5 Identification and Classification

3.5.1 Identification

The minimum identification requirements are detailed in BS EN 60099 - 4, Clause 4.1. This information shall be permanently attached to the arrestor.

3.5.2 Classification

Arrester shall be classified in accordance with BS EN 60099 - 4, Clause 4.2.

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3.6 Construction

3.6.1 Composite Materials

The insulation of surge arrester shall be of a moulded design and be void free. The insulating material will be either Silicon Rubber or EVA with a preference towards a mould in place process.

Composite materials used in the manufacture of insulators shall be traceable from the raw material supplier through the manufacturing process as required under quality assurance procedures.

The insulating material will be described by the manufacturer in defined terms. The use of a generic term such as silicone rubber is not acceptable.

3.6.2 Housing and Sheds

The housing is the external insulating part of an insulator which provides the necessary creepage and protects the internal components from the weather.

Due to the variations in the make up of silicon rubber materials and the potential effects this may have on the long term performance of such a material to provide the following properties:

- UV resistance
- Tracking resistance
- Hydrophobic

The table below provides a good practice guide relating to the composite material content.

Material Composition	Minimum % per weight
Silicone Polymer (Polydimethylsiloxane) (PDMS)	30%
Filler (ATH Aluminium tri hydrate and fumed silica)	64%
Additional parts (pigments, crosslinkers)	1%

Where manufactures differ from this characteristic composition, they shall provide supporting evidence about the long term experience of their product formulation.

The following table provides a list of characteristic properties that result from the above formulation. Manufacturers shall provide details of their characteristic properties for consideration.

Property	Minimum Value
Density	1.5g/ccm
Passing Voltage level of IEC 60587	4.5kV
Flammability class of IEC 60695-11-10 of 3mm specimen	V0
Tensile Strength (Din 53504-S1)	6 N/mm ²
Break Elongation (Din 53504-S1)	300%
UV resistance - @ 300nm the energy of UV wave length equates to a molecular energy breakdown level of 398kJ/mole *	445 kJ/mole

* Assumed wavelength of UV light (sun) 290 – 350nm

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Surge arresters shall be manufactured using the HTV (high temperature vulcanising) and shall ensure that the interface between the housing and the core is chemically bonded. Flash or mould lines shall not exceed 1mm in height.

3.7 Mechanical Loads

Surge arresters may be used as a means of mechanical support and must therefore be suitable for mounting at any angle between the horizontal and vertical. The arresters must therefore satisfy the mechanical loading tests given in BS EN 60099 – 4 and the table below.

	11 kV	20kV	33 kV	66kV	132kV
Cantilever (Nm)	350	350	350	350	350
Torsion (Nm)	50	50	50	50	50
Tension (kN)	2	2	2	2	2

3.8 Short Circuit Failure

Polymeric housed arresters should fail with the housing intact or housing broken down non-explosively in accordance with Clause 8.7 of BS EN 60099 – 4.

3.9 Testing

Gapless metal oxide surge arrestors shall be tested in accordance with BS EN 60099 – 4 with type tests as required in section 8. Routine tests shall be carried out in accordance with BS EN 60099 – 4 Sections 9 and 10.

In addition to standard type tests 100% of arresters shall be tested for internal partial discharge with the arrester energised at 1,05 times the operating voltage (pass result will be $\leq 10\text{pC}$). Also 100% of blocks used in arrester assembly shall be residual voltage tested as required in Section 6.3 of BS EN 60099 - 4. Block sample testing shall also include high current impulse testing at appropriate peak 65kA or 100kA.

Appendix 1 of this specification contains self certification conformance declarations that shall be completed by the supplier for each product offered.

3.10 End of life Disposal

Manufacturers shall provide enough information within Appendix 3 so that all the arrester components may be scrapped and/or recycled in accordance with international and national regulations.

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3.11 Mechanical end fitting requirements

Voltage Type (kV)	Description
11	This Item shall be supplied with M12 stainless steel studs at each end (min length 45mm) complete with 2 flat washers and 2 spring washers per stud. Additionally this item shall be supplied with a flat mounting plate. The arrangement shall be generally in accordance with our drawing 1091480162 sht2
20	This Item shall be supplied with M12 stainless steel studs at each end (min length 45mm) complete with 2 flat washers and 2 spring washers per stud. Additionally this item shall be supplied with a flat mounting plate. The arrangement shall be generally in accordance with our drawing 1091480162 sht3
33 Non Pedestal Mounting	This Item shall be supplied with M16 stainless steel studs at each end (min length 50mm) complete with 2 flat washers and 2 spring washers per stud. Additionally the top stud shall include a line conductor clamp suitable for conductor sizes of 4-35mm diameter. The arrangement shall be generally in accordance with our drawing 1091480165 sht4
33	This Item shall be supplied with M16 stainless steel studs at each end (min length 50mm) complete with 2 flat washers and 2 spring washers per stud. The top stud shall include a line conductor clamp suitable for conductor sizes of 4-35mm diameter. The bottom stud shall be mounted on a pedestal base designed for a 127mm PCD c/w 4xM12 set screws and both flat and spring washers. The base shall be mounted on a set of 4 polyfibre base insulators with 105mm creepage c/w M12 x 20mm deep insert in both ends and M12x 35mm hex head set screws. Additionally the pedestal base shall include a ground connecting plate c/w 2xM10 set screws suitable for the connection of 70mm CSA earthing conductor. The arrangement shall be generally in accordance with our drawing 1091480165 sht3
66	This Item shall be supplied with M16 stainless steel studs at each end (min length 50mm) complete with 2 flat washers and 2 spring washers per stud. The top stud shall include a line conductor clamp suitable for conductor sizes of 4-35mm diameter. The bottom stud shall be mounted on a pedestal base designed for a 127mm PCD c/w 4xM12 set screws and both flat and spring washers. The base shall be mounted on a set of 4 polyfibre base insulators with 105mm creepage and c/w M12 x 20mm deep insert in both ends and M12x 35mm hex head set screws. Additionally the pedestal base shall include a ground connecting plate c/w 2xM10 set screws suitable for the connection of 70mm CSA earthing conductor. This arrangement shall be generally in accordance with our drawing 1091480166 sht1
132	This Item shall be supplied with M16 stainless steel studs at each end (min length 50mm) complete with 2 flat washers and 2 spring washers per stud. The top stud shall include a line conductor clamp suitable for conductor sizes of 4-35mm diameter. The bottom stud shall be mounted on a pedestal base designed for a 280mm PCD c/w 4xM12 set screws and both flat and spring washers. The base shall be mounted on a set of 4 polyfibre base insulators c/w M12 x 20mm deep insert in both ends and M12x 35mm hex head set screws. Additionally the pedestal base shall include a ground connecting plate c/w 2xM10 set screws suitable for the connection of 70mm CSA earthing conductor. This arrangement shall be generally in accordance with our drawing 1091480166 sht3

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4. References

4.1 External Documentation

Reference	Title
BS EN 60099 Part 4: 2006	Surge arresters. Metal-oxide surge arresters without gaps for a.c. systems
ENA TS 43-93 Issue 5: 2004	Line Insulators
IEC 62217 Draft	Polymeric insulators for indoor and outdoor use with a nominal voltage greater than 1000V – general
ENA TS 43-40 Issue 2: 2004	High voltage single circuit overhead lines on wood poles – specification and user guide
ENA TS 41-36 Issue 2: 2004	Distribution switchgear for service up to 36kV (cable and overhead conductor connected)
IEC 60060 – 1: 1990	Guide on high-voltage testing techniques. General
IEC 60587; 2007	Electrical insulating materials used under severe ambient conditions. Test methods for evaluating resistance to tracking and erosion
IEC 60695 – 10 – 11; 2008	Fire hazard testing. Guidance for assessing the fire hazard of electro technical products. Ignitability. Summary and relevance of test methods
IEC 60060 – 1: 1989	Guide on high-voltage testing techniques. General

4.2 Internal documentation

Reference	Title
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5. Definitions

Term	Definition
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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
Steven Salkeld	Standards Engineer	Steven Salkeld	15/06/2009

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
Mark Nicholson	Head of System Strategy	Mark Nicholson	31/07/2009

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
Ged Hammel	Overhead Standards Manager	Ged Hammel	28/07/2009

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
Sean Johnson	DBD Administrator	Sean Johnson	17/08/2009

6.5 Approval

Approval is given for the content of this document

		Sign	Date
Chris Holdsworth	Standards and Assessments Manager	Chris Holdsworth	28/07/2009

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
Carol Morgan	Head of Procurement	Carol Morgan	28/07/2009

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Appendix 1 - SELF CERTIFICATION CONFORMANCE DECLARATION

Gapless Metal Oxide Surge Arrestors are covered by BS EN 60099 – 4:2004 and shall comply with the latest issues of the relevant national and international Standards.

BS EN 60099 - 4 is intended to amplify and/or clarify the requirements of those Standards.

This check sheet identifies the clauses in BS EN 60099 -2 and the clauses of the aforementioned Standards relevant to Gapless Metal Oxide Surge Arrestors for use on current the CE Electric UK distribution network.

The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.

Conformance declaration codes

N/A = Clause is not applicable/ appropriate to the product

Cs1 = The product conforms fully with the requirements of this clause

Cs2 = The product conforms partially with the requirements of this clause

Cs3 = The product does not conform to the requirements of this clause

Cs4 = The product does not currently conform to the requirements of this clause, but the manufacturer proposes to modify and test the product in order to conform.

Instructions for completion

- When Cs1 code is entered no remark is necessary
- When any other code is entered the reason for non-conformance shall be entered
- Prefix each remark with the relevant 'BS EN' or 'ENATS' as appropriate

Manufacturer:

Product Reference:

Name:

Signature:

Date:

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Appendix 1 Table

Gapless Metal Oxide Surge Arrestors

• BS EN 60099 – 4: 2004			
Clause/Sub-clause	Requirement	Conformance Code	Remarks
Section 4.1	Arrestor Identification		
Section 4.2	Arrestor Classification		
Section 5.0	Standard Ratings and Service Conditions		
Section 7.3	Test Samples		
Section 8.0	Type Tests		
Section 9.0	Routine Tests		
Section 10.0	Tests for Polymer-housed Surge Arrestors		
• NPS/001/008 – Technical Specification for Gapless Metal Oxide Surge Arrestors (CE-Electric UK Specification)			
Clause/Sub-clause	Requirement	Conformance Code	Remarks
Section 3.3	Compliance with System Parameters		
Section 3.4	Arrestor Performance Requirements		
Section 3.6.1	Pollution Levels		
Section 3.6.2	Insulation Type		
Section 3.6.3	Insulation Material Composition		
Section 3.7	Mechanical Loads Requirements		
Section 3.9	Additional Testing Requirements		

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Appendix 2 – Technical Particulars

The template below shall be completed for each voltage or design variant offered with appropriate type test evidence, drawings and data sheets submitted.

Manufacturer	
Type and Reference	
Line Discharge Class (BS EN 60099 – 4)	
Nominal Discharge Current (kA)	
High Current Impulse (kA)	
Rated Voltage (kV)	
Continuous Voltage (kV)	
Impulse Voltage (kV) – 1,25/50 us	
Over voltage Capability (kV for 1 second)	
Rated Short Circuit Current – Outdoor Application (kA)	
Power Frequency Voltage – Wet Withstand (kV)	
Flashover Distance (mm)	
Creepage Distance (mm)	
Overall Length (mm)	
Unit Weight (kg)	
Packaging Type	
Cantilever (Nm)	
Torsion (Nm)	
Tension (kN)	
Country of Origin – Arrester	
Country of Origin – Zinc Oxide Blocks	

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Appendix 3 – End of Life Disposal – Material Schedule

The template below shall be completed to identify each discrete component used in the design of the surge arrestor, together with it's material type and any special disposal requirements.

Description of Arrestor Component	Material type (special disposal requirements)